

Volatility Forecasting Engine

Technical Document

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I. Executive Summary - Volatility Forecasting Engine

Volatility sits at the center of options trading. Rather than trying to predict the direction of a market, an approach that is often unstable and difficult to repeat, volatility trading focuses on the magnitude of price movement. This makes it attractive for systematic strategies. Volatility tends to exhibit more persistent structure than raw price levels, responds to macro and regime conditions in measurable ways and can provide opportunities even when markets move sideways.

By forecasting how much an asset is likely to move, rather than which way it will move, traders can construct positions that benefit from realised fluctuations, changes in implied volatility, or discrepancies between the two. Understanding and anticipating volatility therefore gives a more stable foundation for risk management, hedging decisions and extracting value from option markets.

Realised Volatility (RV)

Realised Volatility represents the actual level of volatility that will occur over the coming days. It is calculated from historical price changes, typically using log returns.

This is the volatility a trader experiences and can monetise directly. If RV turns out to be high, a long-gamma trader can profit by dynamically hedging, buying low, selling high and repeatedly capturing convexity.

Implied Volatility (IV)

Implied Volatility reflects the market's expectation of future volatility and is backed out from option prices using a model such as Black-Scholes.

IV is not a forecast of direction; it is a forecast of movement.

When IV is high, options are expensive; when IV is low, they are cheap.

If a trader holds an option and IV rises, the option's value increases, even if the underlying price does not move because the market is now pricing in greater future uncertainty.

How Volatility Is Traded

Long Gamma

A trader buys options (typically straddles/strangles) and benefits when RV is high relative to IV. More realised movement → more profitable dynamic hedging → gamma scalping yields.

Short Vega

A trader sells options when IV is high relative to expected RV.

If realised movement ends up lower than the implied level, the position earns the premium.

Vega Exposure from IV Changes

Even without large price moves, an increase in implied volatility raises the value of long-option positions. Conversely, a decrease in IV benefits short-vega positions.

In every case, the key variable is the relationship between future realised volatility and current implied volatility.

How Our Forecasting Engine Supports These Decisions

To help identify when options are mispriced, our system produces short-horizon RV forecasts using:

- Multi-source daily datasets (prices, volatility indices, yields, FX, commodities, event timing).
- Engineered features capturing microstructure, momentum, volatility compression and regime behaviour.
- A hybrid modelling architecture combining LSTM sequence learning, XGBoost residual modelling and HAR-style components.
- Walk-forward validation to replicate real-time evaluation.

The engine outputs a calibrated estimate of future realised volatility and a comparison against the implied volatility currently embedded in option prices. This enables more disciplined, data-driven long-gamma and short-vega decisions.

II. Background - Options, Volatility & Market Microstructure

Volatility captures how much an asset's price moves and is fundamental to how options behave. Realised volatility tells us what happens; implied volatility tells us what the market expects. When these diverge, opportunities arise. Long-volatility positions can monetise realised movement through gamma scalping, while short-volatility strategies earn risk premia when markets are stable. Understanding delta, gamma, vega and theta from first principles, how option prices respond to price changes, curvature, volatility and time is essential for recognising when options are mispriced. Volatility forecasting sits at the centre of volatility arbitrage. Estimate future RV, compare with IV, identify the volatility regime and adjust long-gamma/short-vega exposures accordingly.

What Is Volatility?

Volatility is the scale of variation in an asset's price. At first principles, it measures how much the price moves, not the direction of the move.

Historical Volatility (HV)

Historical volatility is computed directly from past returns. It is also called Realised Volatility when used ex post.

Historical Volatility (HV) and Realised Volatility (RV) share the exact same formula:

$$HV_n = RV_n = \sqrt{\frac{252}{n} \sum r_t^2}$$

Where:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Both measure the variability of returns over a past window of length n .

The difference is context, not calculation:

Term	Meaning	When used
HV	Vol computed on past data	Used for descriptive/statistical analysis
RV	Vol computed on a defined realised path	Used to test a trading strategy or measure hedging outcome

They are mathematically identical; RV emphasises that we are evaluating the actual vol that occurred over a specific period relevant to P&L.

Computation:

- Take log returns: $r_t = \ln(P_t/P_{t-1})$
- Square them to capture magnitude
- Average the squared returns
- Take the square root
- Annualise with $\sqrt{252}$

This yields a statistical estimate of how noisy the asset has been.

Realised vs Implied Volatility

Realised Volatility (RV)

"What actually happened."

The volatility that determines whether a long-gamma position makes or loses money from hedging.

Implied Volatility (IV)

“What the market thinks will happen.”

Implied from option prices; represents traders’ expectations of future uncertainty.

The IV–RV spread is the foundation of volatility arbitrage.

Vol-of-Vol & Term Structure

Vol-of-vol

Volatility moves over time, sometimes more violently than prices themselves.

High vol-of-vol = market uncertainty changes rapidly.

Term structure of volatility

$$\sigma_{IV}(T_1) \neq \sigma_{IV}(T_2)$$

In calm markets, longer-dated IV > short-dated IV.

Around events (earnings, CPI, FOMC), this can invert sharply.

Why Volatility Is Mean-Reverting

Volatility tends not to stay extremely low or extremely high for extended periods. Instead, it fluctuates around a typical long-run level. When markets become unusually calm or unusually stressed, a combination of information flow, liquidity dynamics and trading behaviour pushes volatility back toward more normal conditions.

- Very low volatility rarely persists, sooner or later new information, events or shifts in sentiment increase market movement.
- Very high volatility usually moderates, as conditions stabilise, liquidity improves, spreads tighten and extreme price swings fade.
- Hedging and trading flows counteract extremes, option hedging, portfolio rebalancing and risk management activity tend to pull volatility back toward more typical levels.

In practice, volatility moves in clusters, but it oscillates around a long-run average rather than drifting indefinitely.

What Are Options?

An option gives the right but not the obligation, to buy or sell an underlying asset at a fixed strike price K before or at maturity T .

- Calls benefit when price rises.
- Puts benefit when price falls.
- Option values depend not only on price but on movement, time and volatility, hence the Greeks.

Delta

- The option’s directional sensitivity.
- Delta is the slope of option value vs underlying price.
- A delta of 0.50 means the option behaves like half a share.

Gamma

- The curvature of the option value.
- Gamma measures how quickly delta changes when the price moves.
- High gamma → deltas move fast → lots of hedging → potential for realised-vol harvesting.

Vega

- Sensitivity to implied volatility.
- A long-vega position benefits when the market prices more uncertainty (IV $\uparrow$).
- Short-vega collects the volatility risk premium but is exposed to volatility spikes.

Theta

- Time decay.
- Options lose value as time passes (all else equal).
- Long options bleed, short options earn theta, unless volatility surprises overwhelm the decay.

Why Long Gamma Is Convexity

From a geometric first principle:

- A curve always lies above its tangent line.
- Gamma is the curvature.
- When you dynamically hedge, you repeatedly buy low and sell high because your delta overshoots and undershoots the actual curve.

This convex payoff is why long gamma can monetise realised volatility.

Why Short Vega Collects a Risk Premium

Markets typically price volatility with a cushion, a premium for uncertainty, jumps and tail risks.

Selling vega earns this premium until volatility spikes.

What Is Gamma Scalping?

Gamma scalping is the practical expression of convexity.

It is the process of continuously delta-hedging a long-gamma option to monetise realised volatility.

Intuition

- Price goes up $\rightarrow$ delta becomes positive $\rightarrow$ you sell underlying to rebalance.
- Price goes down $\rightarrow$ delta becomes negative $\rightarrow$ you buy underlying to rebalance.
- These flows create a systematic buy low, sell high pattern.

Your hedges generate P&L proportional to how much realised volatility occurs, scaled by your gamma.

Gamma Scalping as “Energy Harvesting” - Anecdotal Explanation

One useful way to think about gamma scalping is to imagine you are operating a wave-energy turbine on the coastline:

- Calm waters $\rightarrow$ little energy generated.
- A steady swell $\rightarrow$ small but consistent energy flow.
- A choppy sea with lots of small waves $\rightarrow$ high energy harvested.

Your option's gamma is the turbine.

The market's realised volatility is the motion of the waves.

Your delta hedges are the mechanism that captures the energy each time the water rises or falls.

As long as:

- The waves keep coming (RV), and
- The maintenance cost of running the turbine (theta) is not too high,

you profit from movement itself.

This anecdote maps directly onto the real mechanics of gamma scalping, movement (RV) is the raw energy, convexity (gamma) converts it into P&L, and hedging is how you capture it.

When Gamma Scalping Works

- $RV > IV$ (classic)
- Markets whip around inside a range
- High-frequency noise, uncertain news flow, liquidity pockets

Your long gamma earns enough through hedging to overcome theta decay.

When Gamma Scalping Fails

- IV spikes sharply: your long options may gain but hedging losses can overwhelm if you are short vega or under-hedged.
- Macro events: non-linear jumps break the smooth hedging assumption.
- Flat, low-vol markets: not enough realised movement to offset theta.

The engine's purpose is to avoid these failure regimes by quantifying risk.

What Is Volatility Arbitrage?

Volatility arbitrage exploits mispricings between implied volatility (what options cost) and expected realised volatility (what your model forecasts).

The Central Spread: $IV - \text{Forecast}(RV)$

Principle:

- If $IV > \text{forecast}(RV) \rightarrow$ options are expensive $\rightarrow$ opportunity to sell volatility
- If $IV < \text{forecast}(RV) \rightarrow$ options are cheap $\rightarrow$ opportunity to buy volatility

Your forecasting engine provides the fair value of future RV.

Long-Vol vs Short-Vol Opportunities

- Long-vol / long gamma:
Buy when options are cheap or regimes are fragile (RV likely to rise).
- Short-vol / short vega:
Sell when volatility is overpriced relative to stable conditions.

Regime Dependency

The behaviour of volatility is not stationary.

Different conditions support different strategies:

- Calm regime:
Mean reversion dominates $\rightarrow$ short volatility earns risk premia.
- Transitional or turbulent regime:
RV tends to rise $\rightarrow$ long gamma positions can outperform.

The purpose of the pipeline is to detect the regime, forecast RV, and assign reliability metrics for each signal.

III. High-Level Architecture of the Full Volatility Forecasting Pipeline

This section provides a systems-level overview of the full volatility forecasting and signal-generation engine. Its goal is to orient the reader before diving into individual components, how the pieces fit together and why each layer is necessary for accuracy, robustness and real-world trading use.

The pipeline follows a seven-layer architecture, progressing cleanly from raw market data → validated features → model outputs → tradeable signals.

Every layer is deterministic, reproducible and fully auditable, ensuring institutional-grade reliability.

1. Data Ingestion Layer

The engine begins by constructing a unified, cross-asset dataset from all relevant volatility-drivers:

- Equity ETFs (QQQ, SPY, IWM, sector ETFs)
- Index-level implied volatility metrics (VIX, VXN, RVX, sector vol indices)
- Realised volatility windows (RV_{14} , RV_{21} , RV_{63})
- Macro and cross-asset data (MOVE index, FX majors, credit spreads, yields, commodities)
- Derived technical features (ATR%, Bollinger bandwidth, price entropy, vol-of-vol, autocorrelation, compression metrics)

All inputs pass through a unified timestamp normaliser that converts every row into UTC-consistent YYYY-MM-DD dates, ensuring that cross-ticker alignment is mathematically safe for downstream modelling.

This forms the foundation of the entire stack.

2. Institutional-Grade Data Validation Layer

Before any modelling occurs, the system verifies that the data is internally coherent and cross-asset consistent.

Validation runs in two stages:

(i) Per-Ticker Validation

Ensures each asset's time series is structurally sound:

- Calendar continuity (no date jumps)
- OHLC geometry checks
- Return sanity filters and gap detection
- Macro/FX outlier filters
- RV/IV anomalies via rolling z-scores
- Missingness and stale-data analysis

This produces a per-ticker `health_score`, enabling filtering of structurally weak assets.

(ii) Global Unified Validation

Ensures the entire universe is jointly consistent:

- Duplicate (symbol, date) detection
- Cross-ticker history alignment
- Universe-wide NaN-flood detection
- Cross-asset volatility sanity checks

The result is a clean, institution-grade dataset ready for modelling.

3. Feature Pruning & Economic Alignment Layer

The raw dataset contains more than 200 potential explanatory variables.

The engine applies a multi-stage pruning pipeline that prioritises interpretability, stability and economic validity:

- Missingness and zero-variance filtering
- High-correlation clustering (remove redundancy)
- XGBoost gain scoring
- Linear coefficient sign checks
- Economic-sign alignment (e.g., if bandwidth increases → RV should increase)

Each feature receives a Feature Quality Score (0–100).

Only robust, economically interpretable features reach the modelling layer, ensuring that the models learn meaningful relationships rather than noise.

4. Universe Eligibility Scanner

Not all tickers are model-ready. A ticker must satisfy strict quality criteria before being admitted into training or forecasting:

- Covariance stability (Ledoit–Wolf shrinkage)
- Per-ticker missingness thresholds
- LSTM sequence viability (minimum lookback coverage)
- Forward-target availability
- Feature completeness in the latest row

This prevents unstable assets, data gaps or structural inconsistencies from contaminating the global model.

5. Global Forecasting Engine (XGBoost + LSTM)

At the heart of the pipeline sits a dual-model ensemble:

XGBoost

- Learns cross-sectional drivers of future volatility
- Fast, interpretable and highly sensitive to feature importance
- Generates P10/P50/P90 forecasts and model-specific uncertainty

LSTM

- Captures sequence structure (memory, regimes, slow trends)
- Adapts to changing market conditions more smoothly
- Particularly strong during regime shifts or clustering phases

Both models forecast 14-day realised volatility, each with its own view of risk and uncertainty.

Their outputs are later combined via a regime-weighted ensemble, increasing robustness across environments.

6. Event & Regime Layer

Volatility is fundamentally event-driven, so the engine enriches raw forecasts with a set of event and regime-sensitive metrics:

- Proximity to macro catalysts (CPI, NFP, FOMC, PPI)
- Structural volatility events (OPEX, VIX expiry, ETF rebalance)
- Exponential event-pressure scores
- Drift metrics (Mahalanobis regime deviation)
- Walk-forward validation (WFA) confidence
- Regime similarity confidence

These metrics guide the ensemble weights and help identify whether the current market is high-risk, stable, compressed or transitioning.

7. Signal Engine & Execution Layer

The final layer converts enriched forecasts into actionable, trade-ready signals:

- Long Gamma Entry (when $RV > IV$ expectations + favourable regime)
- Short Vega Entry (selectively, when IV is overpricing risk)
- Gamma Exit Conditions (IV collapse, regime shift, event risk)
- Volatility Regime Classification (calm, compressed, trending, storm)
- Diversification Tags (cross-ticker correlation mapping)
- Trade Ranking (IV-RV edge, reliability-adjusted)

Outputs are compiled into a clean CSV designed for discretionary review or automated routing into an execution system.

IV. Data Preparation & Pre-Modelling Processing

This chapter explains the full data preparation pipeline that transforms raw multi-source market data into a stable, modelling-ready dataset. In a volatility forecasting system, data quality is not an administrative concern, it is a core predictive asset. Even small structural inconsistencies, missing data or regime breaks can overwhelm any model's signal, especially in a domain as sensitive as volatility.

The preparation pipeline is therefore engineered around three principles:

- Quality: remove errors before they propagate
- Robustness: produce a consistent dataset across tickers and time
- Economic integrity: ensure features behave as markets logically dictate

The result is a clean, harmonised dataset that the forecasting engine can rely on.

1. Raw Data Inputs & Derived Features

The system ingests multi-source data for every ticker and constructs a rich set of price-based, implied-volatility and macro/cross-asset features. All rolling metrics are built deterministically so the pipeline remains reproducible and auditable across reruns.

Price-Based Features

- OHLCV bars
- Log returns
- Gap percentage
- Rolling realised volatility windows (RV_{14} , RV_{21} , RV_{63} ...)
- Parkinson / Garman–Klass volatility
- ATR & ATR%
- Bollinger bandwidth
- Entropy & return-dispersion statistics

These capture microstructure, trend, noise structure, and compression signals relevant to forward volatility.

Implied Volatility Features

- ATM IV (14D, 30D)
- IV momentum & slope
- IV–RV spread
- IV percentile rank
- Vol-of-IV (IV volatility)

These provide a forward-looking view of market pricing and expectations.

Macro & Cross-Asset Features

- VIX, VXN (equity vol surfaces)
- MOVE index (rates volatility)
- FX majors (EURUSD, USDJPY, GBPUSD)
- Credit spreads / HYG
- Key commodity vol proxies

These help the model capture cross-asset spillover effects that strongly influence equity volatility regimes.

2. Data Validation (Per-Ticker & Global)

After feature construction, the data-validation module ensures the dataset is both internally consistent per ticker and externally coherent across the entire universe.

Per-Ticker Validation

Each asset undergoes a battery of structural tests:

- Calendar integrity: missing or duplicate dates trigger warnings or hard failures
- Candle geometry: OHLC must satisfy market constraints
- Return & gap sanity: abnormal 20%+ returns or 15%+ overnight gaps are flagged
- Volatility anomalies: rolling z-scores for RV, IV, and volume
- Macro-signal sanity: VIX, MOVE, and FX values must fall within realistic bounds

Per-ticker results feed into a health score, enabling filtering of unstable assets.

Global Validation

The system then verifies that all tickers together form a coherent financial universe:

- Symbol and date duplication detection
- Cross-ticker date-range alignment
- Symbol coverage tests
- Global missingness & NaN-flood detection
- Cross-ticker RV/IV divergence testing
- Target leakage detection

The global validation report summarises the system's readiness for modelling.

3. Feature Pruning & Dimensionality Reduction

With >200 initial features, reducing the space to a robust and interpretable subset is essential. This is achieved through a multi-stage pruning pipeline.

Missingness & Structural Filters

Features are removed if they exhibit:

- 40% missingness
- Zero variance
- Too few unique values

This step eliminates structurally weak variables.

Correlation Clustering

Highly correlated variables ($|\rho| > 0.92$) are grouped and reduced to a single representative.

This prevents redundancy, overfitting, and unstable covariance matrices.

Predictive Utility (XGBoost Gain Score)

A lightweight XGB model ranks features by predictive contribution.

This identifies variables with genuine out-of-sample utility.

Linear Explainability Checks

A linear regression provides:

- Coefficient sign
- Magnitude
- Direct correlation with the target

This ensures the feature behaves in a stable, interpretable way.

Economic-Sign Alignment

Every feature must exhibit a directionally sensible relationship with future realised volatility.

Examples:

- ATR $\uparrow \rightarrow$ future RV expected $\uparrow$ (positive)
- RSI $\uparrow \rightarrow$ future RV typically $\downarrow$ (negative)

Features that violate economic intuition are rejected.

Feature Quality Score (FQS)

A composite FQS blends:

- XGB gain
- Standardised linear coefficients
- Target correlation
- Economic-sign alignment

Final features are selected via:

- Top-N ranking, and/or
- $FQS \geq \text{threshold}$

Only the highest-quality features proceed.

4. Ticker Eligibility & Universe Filtering

Before modelling begins, each ticker must pass a series of viability tests.

Target Availability

The ticker must contain enough future RV_{14} observations to support supervised learning.

Missingness Thresholds

Overall feature completeness must exceed 90%, limiting the need for imputation.

Sequence Viability (LSTM Requirements)

A ticker must supply at least:

- 22 consecutive non-missing rows (for a stable LSTM sequence window)

Covariance Stability

Using Ledoit–Wolf shrinkage, the ticker’s feature covariance matrix is tested for:

- Invertibility
- Reasonable conditioning

This prevents unstable covariance structures from propagating into models.

Latest-Row Completeness

Live forecasting requires the most recent row to be fully populated.

Tickers failing this are excluded from the active universe.

End Result of Data Preparation

The output of this chapter is a clean, stable, economically coherent, cross-sectionally harmonised dataset that forms the foundation of the entire forecasting pipeline.

High-quality data preparation ensures that:

- Forecasts are reliable,
- Signals are trustworthy, and
- Volatility arbitrage strategies operate on true market information, not artefacts.

This prepared dataset is the backbone of every forecast, model decision and trading signal that follows.

V. Model Architecture (XGBoost + LSTM + Regime-Weighted Ensemble)

This chapter describes the forecasting engine responsible for predicting 14-day forward realised volatility (RV₁₄ forward).

The design follows a hybrid modelling philosophy, combining:

- XGBoost, which excels at cross-sectional nonlinear tabular structure.
- LSTM networks, which excel at learning sequential dynamics and volatility regimes.
- A regime-aware ensemble, which blends the two adaptively using statistical confidence measures.

Before detailing the architecture, we first explain how each model works mechanically so readers understand why these tools are used in volatility forecasting.

How the Models Work

XGBoost - A System of Many Competing Decision Rules

From first principles, XGBoost is:

- A large collection of small decision trees
- Each tree corrects the mistakes of the previous tree
- Uses gradient descent on a loss function

Mechanically:

- The first tree makes coarse predictions (e.g., “vol goes up when bandwidth is high”).
- The next tree focuses on the errors made by the first tree.
- The following tree focuses on remaining errors.
- This continues for hundreds of small trees.

Each tree is weak, but together they form a powerful nonlinear function:

$$\hat{y}(x) = \sum_{m=1}^M f_m(x)$$

XGBoost also:

- Performs automatic interaction detection (e.g., ATR% × VIX × trend)
- Handles nonlinear boundaries gracefully
- Is robust to messy, correlated financial features
- Works extremely well cross-sectionally
- Provides feature importance and explains why forecasts changed

In volatility forecasting, XGBoost is ideal for detecting structural relationships across tickers, macro data, and technical indicators.

LSTM - A Memory System for Sequences

An LSTM (Long Short-Term Memory network) is fundamentally a learnable memory system.

Mechanically, it works as follows:

- The LSTM reads one day of features at a time.
- It decides what information to store, forget, or output.
- It builds an internal “state” that evolves over time.
- This state captures long-memory patterns such as volatility clustering.

The three gates inside each LSTM cell:

- Forget gate → erase irrelevant past information
- Input gate → decide what new information to store
- Output gate → decide what to emit to the next step

This structure makes LSTMs excellent at identifying:

- Regime persistence
- Sequential transitions
- Volatility clustering
- Slow-moving macro conditions
- Pre-event tension or compression

Where XGBoost reads rows independently, LSTM understands how yesterday teaches us about today.

Why Combine Them?

Volatility is both:

- Cross-sectionally structural (macro, IV surfaces, features), and
- Sequentially dynamic (clustering, memory, regimes).

A single model class cannot capture both behaviours.

The hybrid architecture leverages the strengths of each.

Target Definition - 14-Day Forward Realised Volatility

The model predicts:

RV14 forward, ie realised volatility over the next 14 trading days.

This target is chosen because:

- It aligns with common listed-option maturities (1–2 weeks).
- Short-term RV has predictable clustering behaviour.
- This horizon is ideal for gamma scalping and vol-arb timing.
- It provides a clean, forward-looking measure without look-ahead bias.

All signals and IV-RV comparisons revolve around this forward volatility target.

Model 1 - XGBoost (Cross-Sectional Gradient-Boosted Trees)

Role in the Ensemble

XGBoost contributes:

- Cross-sectional nonlinear structure
- Robustness to multicollinearity
- Fast adaptation to regime shifts (trees split on regime boundaries)
- High interpretability via feature importance

It is especially strong in noisy or discontinuous markets where threshold-based relationships dominate.

Input Features

XGB consumes a compact set of top-ranked features:

- Vol clustering (RV_1 , RV_7 , RV_{21} ...)
- IV structure and IV–RV spreads
- ATR%, entropy, bandwidth, jump flags
- Macro vol indices (VIX, VXN, MOVE)
- Cross-asset risk signals (HYG, DXY, FX vol)
- Trend and range proxies

All features are median-imputed and standardised.

Walk-Forward Validation (WFA)

XGB is trained under an expanding-window WFA procedure:

- 14-day forward test window
- Training window grows with time
- WFA confidence stored per ticker
- Prevents leakage and ensures live realism

This produces time-consistent, out-of-sample-robust predictions.

Model 2 - LSTM (Sequence-Based Volatility Dynamics)

Role in the Ensemble

LSTMs add temporal intelligence:

- Detect volatility clustering
- Identify transition phases
- Recognise slow-moving structural regimes
- Understand temporal lead-lag effects in features

This complements XGB's tabular strengths.

Input Structure

The LSTM uses:

- Sliding windows of 22 days (one trading month)
- Pruned, normalised features
- Strict ticker eligibility to avoid gaps or instability

Architecture

A lightweight but stable setup:

- 1–2 LSTM layers (32–64 units)
- Dropout regularisation
- Dense output layer predicting $RV_{14_forward}$

Designed for speed and low overfitting risk.

Regularisation & Stability

To ensure realism and reproducibility:

- Early stopping on validation MAE
- RobustScaler normalisation
- Fixed seeds for determinism
- Conservative dropout tuning

LSTMs excel during trending regimes or periods with strong memory effects.

Forecast Combination - Regime-Aware Ensemble Layer

The final forecast is not an average.

It is a dynamic blend driven by regime diagnostics.

Each model outputs:

- P10 / P50 / P90
- Uncertainty metrics
- Confidence scores (WFA, drift, similarity)

Inputs to Weighting

Ensemble weights depend on:

- Model Reliability Score
- Regime Similarity Score (from Mahalanobis distance)
- WFA Confidence
- Drift Score (how abnormal today is)
- Economic-sign consistency

Weighting Logic

- Stable regimes: XGBoost weighted higher
- Transition regimes / noisy markets: LSTM weighted higher
- Chaotic / event-driven markets: fallback to conservative XGB weighting

This allows the ensemble to adapt automatically to market conditions.

Ensemble Outputs

- Ensemble_Forecast_RV_14
- Ensemble_p10 / Ensemble_p90
- Forecast Spread (p90 – p10)
- Ensemble_vs_IV (key vol-arb edge metric)

These feed directly into the signal engine.

Quantifying Forecast Uncertainty

Because volatility is inherently uncertain:

- The ensemble outputs uncertainty bands
- Wide spreads → avoid trades
- Narrow spreads → valid signals
- Confidence gating prevents false positives in noisy markets

This ensures risk-controlled, realistic tradeability.

Model Interpretability & Diagnostics

To maintain transparency, the engine stores:

- XGB feature importance (gain + split counts)
- LSTM permutation importance
- Correlation snapshots
- Linear coefficient signs
- Regime-drift and reliability diagnostics

This supports debugging, oversight, and communication with risk teams.

Why the Hybrid Architecture Works for Vol-Arb

Volatility is characterised by:

- Regime persistence
- Clustering
- Cross-asset dependencies
- Macro-event sensitivity
- Nonlinear threshold behaviour
- Occasional jumps

A single model cannot capture all these dynamics:

- XGB captures nonlinear cross-sectional behaviours
- LSTM captures sequential and regime dynamics
- Regime-Weighted Ensemble integrates both intelligently

This architecture delivers:

- More stable forecasts
- Stronger IV-RV edge detection
- Cleaner gamma scalping signals
- Reduced noise during macro events
- Better adaptability to different market states

The hybrid system significantly outperforms either model class alone.

VI. Signal Generation & Trading Strategy

This section describes how the forecasting engine transforms raw model outputs into actionable, risk-aware volatility trading signals.

The signal layer combines:

- IV-RV mispricing
- Ensemble uncertainty
- Macro-event proximity
- Volatility regime conditions
- Model Reliability Scores

This allows us to identify when long gamma or short vega opportunities are both present and safe to trade.

The entire framework is intentionally conservative, reflecting institutional risk controls and the reality that gamma scalping fails in unstable or event-driven markets.

The goal is not to maximise signal quantity, but signal quality.

Overview of the Signal Layer

The forecasting engine produces four core components:

- Point forecast:
Ensemble_Forecast_RV_14
- Uncertainty band:
(P10, P50, P90)
- Confidence scores:
WFA confidence, regime similarity, drift, and the combined Model Reliability Score (MRS)
- IV-RV spreads:
IV_14 – Forecast_RV_14

The signal engine consumes these metrics to determine:

- Is the option mispriced?
- Is the forecast trustworthy?
- Is the market stable enough to trade?

Signals focus on 14-day forward volatility, aligning with weekly/bi-weekly listed options and the RV₁₄_forward prediction horizon.

Event Engine Integration

Volatility behaves differently around macroeconomic and structural catalysts.

The event engine module annotates every ticker with:

- Upcoming macro event (CPI, NFP, FOMC, PPI)
- Structural vol events (OPEX, VIX expiry, ETF rebalance)
- Event “pressure” score (exponential decay based on proximity and importance)
- Binary flags: Macro_Window_Live, Vol_Window_Live

These influence trading in three ways.

Hard Blocks During Event Windows

No trades are allowed when:

- A macro window is active
- A vol-structure event is live
- A macro event is within the next 3 days

This prevents taking gamma or vega positions ahead of unpredictable jumps.

Adjusting Signal Conviction

The event engine also influences:

- Diversification tags (macro-driven vs vol-driven)
- Regime classification
- Long gamma/short vega entry decisions

Events often introduce nonlinear, gap-prone behaviour and the signal engine reduces exposure accordingly.

Volatility Regime Classification

The Model Reliability Score (MRS) combines:

- WFA performance
- Regime similarity
- Mahalanobis drift score
- Data-quality and recency metrics

This single value determines the market regime:

Score	Regime	Interpretation
≥ 70	Stable Regime	High predictability, signal trusted
40-70	Mixed Regime	Some uncertainty; reduced weighting
< 40	Chaotic Regime	Avoid signals; jump-driven or unstable

Regime classification affects:

- Ensemble weights
- Entry thresholds
- Exit logic
- Signal validity

In effect, MRS acts as a volatility market weather forecast.

Long Gamma Strategy Logic

A long gamma position is taken when realised volatility is expected to rise, meaning options are undervalued.

Entry Conditions

All six must hold:

Vol Mispricing

1. Undervalued Volatility

$$IV_{14} < \text{Forecast}_{p10}$$

→ Even the lowest bound of expected RV exceeds IV → options are cheap.

High Conviction

2. Low Forecast Uncertainty

$$\text{Forecast_Spread} < 6 \text{ vol pts}$$

→ Narrow p10–p90 band → the ensemble agrees.

3. Upward RV Shift Expected

$$\text{Ensemble_Forecast_RV}_{14} - RV_{14} > 1.0 \text{ vol pts}$$

→ Model expects realised vol to increase.

Favourable Market Structure (Compression Environment)

4. $bb_bandwidth_pctile < 30$

5. $atr_pct_pctile < 35$

→ The market is tightly coiled; compression → expansion dynamics support long gamma.

Model Stability & Risk Controls

6. High Model Reliability

$MRS > 60$

7. No Macro Window Active

→ No FOMC, CPI, earnings or major macro catalysts.

If all conditions are TRUE → Long Gamma Entry = 1.

Short Vega Strategy Logic

Short vega is the mirror case: when volatility is overpriced and expected to fall.

Entry Conditions

All must hold:

Vol Mispricing

1. Rich Volatility

$IV_{14} > Forecast_p90$

→ Even the upper bound of RV is below implied → options are expensive.

Conviction

2. Low Forecast Uncertainty

$Spread < 6$

3. High Reliability

$MRS > 65$

Risk Control

4. No Macro Event Window Active

If all conditions are TRUE → Short Vega Entry = 1.

Long Gamma Exit Conditions

Even a correct entry can become unsafe. A long-gamma position should be exited immediately if any of the following conditions hold.

Exit Triggers (Any one is enough)

1. Expected RV < Current RV
→ RV compression likely → gamma profits drop sharply.
2. IV rises above forecast
 $IV_{14} > \text{Forecast}_{p50}$
→ Option is no longer cheap; edge has decayed.
3. Forecast Uncertainty Widens
Spread > 10
→ Model loses conviction.
4. Macro Event Within 3 Days
→ Avoid gamma into event risk; hedging becomes unreliable.

If any condition is TRUE → End Long Gamma Condition = 1.

IV/RV Spreads & Trade Ranking

All active signals receive:

- Abs_Ensemble_vs_IV (absolute mispricing size)
- Cross-asset ranking (largest mispricing → highest priority)

This provides a clear, execution-ready ordering of arbitrage opportunities.

Diversification Tags

Each signal also receives a high-level environment tag:

- Macro Regime - dominated by macro catalysts
- Vol Regime - driven by structural volatility events
- Neutral - balanced conditions

This prevents clustering risks such as:

- FOMC volatility
 - OPEX / VIX expiry distortions
 - Cross-asset vol spillovers
-

Final Output: Execution-Ready CSV

trade_signals_today.csv contains:

- Symbol
- LongGamma_Entry
- ShortVega_Entry
- End_LongGamma_Condition
- Abs_Ensemble_vs_IV
- Volatility_Regime
- Diversification_Tag

Designed for immediate use in discretionary or automated systems.

Why This Signal Engine Works in Practice

The system is highly selective due to:

- Strong mispricing requirements
- Strict uncertainty gating
- Regime alignment
- Macro event awareness
- Directional RV dynamics
- Practical gamma-scalping considerations

A trade occurs only when:

- A genuine IV–RV mispricing exists
- The model has conviction
- The market structure supports the trade
- Risk environment is stable

This replicates institutional quant-volatility practice, reducing false positives and ensuring only high-quality, high-edge signals.

VII. System Outputs and Dashboarding

This section describes the outputs generated by the volatility forecasting engine and how these are used to monitor model behaviour, interpret forecasts and support trading decisions. All outputs are presented in a transparent, audit-friendly structure, enabling rapid inspection, reliable backtesting and seamless downstream visualisation.

Per-Ticker Dashboard CSVs

For every ticker in the universe, the engine produces a dedicated dashboard file:

vol_dashboard_data_<SYMBOL>.csv

Each file captures the complete state of the model, the market and the event environment at the close of the current trading day.

Forecast Metrics

These fields describe the model's core volatility forecasts:

- Forecast_RV_14 (LSTM P50)
- Forecast_RV_14_P10 / Forecast_RV_14_P90
- XGB_Forecast_RV_14 and P10 / P90
- Ensemble_Forecast_RV_14 (final blended forecast)
- Ensemble_Spread (P90 – P10 uncertainty)
- Ensemble_vs_IV (forecast–IV mispricing)

These allow analysts to immediately see model expectations, the uncertainty envelope, and the magnitude of any IV–RV arbitrage opportunities.

Market State Metrics

A snapshot of the current market environment:

- IV_14, RV_14
- Bollinger Bandwidth Percentile
- ATR% Percentile
- Close price, recent log-returns

These contextual indicators show whether volatility is compressed, expanding or structurally unstable, essential for interpreting long-gamma or short-vega signals.

Model Diagnostic Metrics

Detailed diagnostics supporting transparency and trust in model outputs:

- Ensemble weights (LSTM vs XGB contribution)
- Relative MAE estimates (per-ticker model fit)
- Model disagreement ($|\text{LSTM} - \text{XGB}|$)
- Regime similarity score (pattern matching vs historical regimes)
- Drift score (Mahalanobis distance - deviation from “normal”)
- Model Reliability Score (0–100)

These metrics explain why the model is confident or uncertain, and anchor the institutional-style gating used in the signal engine.

Event Metrics

Indicators showing the influence of macro and vol-structure events:

- Macro window active (True/False)
- Volatility window active (True/False)
- Days to next macro/vol event
- Event pressure scores (severity and magnitude)

These allow users to quickly understand whether event risk is driving model behaviour or suppressing signals.

Summary Metrics File

The engine also produces a consolidated universe-level snapshot:

summary_metrics.csv

This file rolls up all critical forecast, regime, and diagnostic indicators across the full ticker set. It serves as the primary input to:

- The event engine
- The signal engine
- Backtesting & research workflows

Key fields include:

- Latest forecasts (P10/P50/P90)
- IV/RV spreads (absolute and relative)
- Regime classification scores
- Ensemble uncertainty measures
- Feature-level diagnostics

This provides a single, clean table for daily analysis, batch validation, and cross-asset monitoring.

VIII. Evaluation & Risk Controls

This chapter outlines the evaluation and risk-control components that are in the volatility forecasting engine. These components ensure that forecasts and signals are robust, conservative and safe for daily operational use.

Evaluation Framework

The system currently supports the following validation and diagnostic features, all of which are implemented inside the forecast runner and summary-metrics workflow.

Walk-Forward Validation (Per Ticker)

XGBoost module implements expanding-window walk-forward validation, producing:

- Out-of-sample MAE estimates
- Rolling performance statistics
- Confidence penalties used inside the Model Reliability Score (MRS)

Cross-Sectional Diagnostics

Summary CSVs include:

- Regime similarity
- Drift score (Mahalanobis)
- Ensemble disagreement ($|\text{LSTM} - \text{XGB}|$)
- Ensemble spread (p90–p10)

These provide structural, regime-aware evaluation of model behaviour.

Forecast Distribution Checks

Implemented daily metrics include:

- P10 / P50 / P90 volatility forecasts
- Ensemble spread as uncertainty
- IV vs. forecast mispricing (Ensemble_vs_IV)

These allow you to inspect forecast quality visually and via downstream dashboards.

Model Diagnostics

All of the following diagnostics exist today in the output CSVs:

- Model Reliability Score (MRS)
Aggregates drift, WFA stability, regime similarity, spread width, and model disagreement.
- Model disagreement metric
Absolute difference between LSTM and XGB.
- Ensemble weights
Contribution of each model to the final forecast.
- Regime similarity scores
Pattern-matching versus historical volatility regimes.

These diagnostics act as built-in quality checks and support safe signal generation.

Event & Regime Controls

Macro and Volatility Event Engine

Your current event engine provides:

- Macro window flags (CPI, NFP, FOMC)
- Vol-window flags (VIX expiry, OPEX)
- Days to next event
- Event pressure scores

All of these features are already used inside the signal gating logic.

Regime Classification

Your system tags each ticker/day as:

- Macro regime
- Vol regime
- Neutral

This supports diversification and prevents clustering risk.

Signal Gating Risk Controls

The current signal engine applies the following coded risk controls:

Event Blackouts

Signals are blocked when:

- A macro window is active
- A vol-structure event is active
- A macro event is within 3 days (gamma-exit logic)

Uncertainty Filters

No long-gamma or short-vega signals are generated unless:

- Ensemble Spread < threshold

This ensures signals occur only when uncertainty is low.

Reliability Score Thresholds

Signals require:

- MRS > 60 (long gamma)
- MRS > 65 (short vega)

These thresholds are already embedded in your signal logic.

Compression Filters

Long gamma requires:

- bb_bandwidth_pctile below threshold
- atr_pct_pctile below threshold

IX. Limitations & Model Risk

Every quantitative volatility model has structural limitations and inherent risks. This section documents these transparently to support realistic expectations and robust discussions with stakeholders.

Volatility Is Not Fully Predictable

Even the most advanced models cannot fully anticipate future volatility because:

- Market volatility is driven by news shocks, liquidity breaks, and macro announcements.
- Regime shifts can occur rapidly and non-linearly.
- Sudden jumps (earnings, geopolitical surprises) can invalidate short-term statistical patterns.

The engine captures structural behaviour well but volatility itself remains inherently stochastic and partially unpredictable.

Feature Drift & Data Quality Risk

Certain indicators evolve over time:

- Measures such as Bollinger Bandwidth or ATR% can change their behavioural meaning as market microstructure evolves.
- Data quality inconsistencies (missing values, delayed updates, provider errors) can introduce noise into forecasts.
- Feature drift may temporarily weaken model reliability until recalibration or retraining occurs.

Your pipeline mitigates these risks through robust preprocessing but they cannot be eliminated entirely.

Event Risk Remains Partially Unmodelled

The event engine blocks trades around known macro releases, but:

- Some events have nonlinear, second-order impacts that extend beyond predefined windows.
- Political shocks, unexpected headlines, and liquidity vacuums are fundamentally unpredictable.
- Volatility microstructure around FOMC or CPI can show whipsaw dynamics not captured by daily forecasts.

No volatility model can perfectly adjust for all event-driven uncertainty.

Model Stack Complexity

The hybrid architecture (LSTM + XGB + Ensemble) improves robustness but also introduces model-stack risks:

- Ensemble weighting may behave sub-optimally in extremely chaotic or rare regimes.
- LSTM may overfit subtle temporal noise if the data regime shifts.
- XGBoost may overweight spurious cross-sectional patterns during unusual market conditions.
- Model disagreement (LSTM vs XGB) can rise unexpectedly, reducing confidence.

These risks are partly mitigated via the Model Reliability Score but cannot be eliminated.

Execution Risk

Signal correctness does not guarantee realised P&L:

- Slippage, wide bid-ask spreads and execution timing can erode expected edge.
- Gamma scalping relies on stable liquidity, thin markets reduce hedge efficiency.
- Availability of stock for hedging may vary during stressed market conditions.
- IV-RV edges may be correct but difficult to monetise if trading costs spike.

Execution risk is an unavoidable component of real-world volatility trading.

Parameter Sensitivity

Several design choices influence model behaviour:

- The fixed 14-day horizon may not always align perfectly with the optimal gamma scalping window for every ticker or regime.
- Ensemble weighting introduces sensitivity to regime scores and disagreement metrics.
- Thresholds for uncertainty, compression and reliability scores reflect conservative intuition but may require periodic recalibration.

These sensitivities highlight the importance of ongoing monitoring and version-controlled iteration.

Summary

This chapter demonstrates clear awareness of:

- Model uncertainty
- Event-driven failure modes
- Regime instability
- Execution constraints
- Parameter sensitivity
- Data and feature drift

These limitations are normal in institutional volatility modelling and show that the system is designed with realistic expectations and responsible risk framing.

X. Future Improvements & Research Roadmap

This section outlines the next set of enhancements that would materially improve forecasting accuracy, signal robustness, and practical trading performance. These items represent a forward-looking roadmap for turning the current forecasting and signal engine into a fully institutional quant-volatility platform.

Advanced Machine Learning

- Transformers / Temporal Fusion Transformers (TFT): Better long-range pattern capture and interpretability.
- Graph-based volatility models: Link cross-asset structures (VIX, sectors, indices).

Bayesian Uncertainty & Scenarios

- Bayesian neural networks for more reliable P10–P90 intervals.
- Variational inference (SVI) for scalable uncertainty.
- Scenario engines to simulate macro or liquidity shocks.

Macro-Aware Modelling

- Incorporate Fed funds curve, inflation surprises and yield-curve factor transitions to improve regime detection.

Microstructure Features

- Add intraday RV, order book imbalance, VPIN and liquidity metrics to refine short-term gamma signals.

Option Chain Integration

- Full IV surface ingestion, skew/SABR features and term-structure shock modelling for deeper understanding of implied vol dynamics.

Full Backtesting & Performance Engine

- Directional metrics: Hit Rate, Directional Accuracy.
- Economic metrics: IV–RV edge realisation; long-gamma / short-vega P&L proxies.
- Portfolio backtesting: exposure clustering, diversification effects.
- Execution simulation: hedging costs, slippage, transaction costs.

Portfolio Optimisation

- Exposure balancing across correlated tickers.
- Position sizing linked to model reliability and regime.
- Adaptive leverage based on volatility state.

Infrastructure & Scaling

- Orchestration: Airflow/Prefect for automated daily runs.
- Containerisation: Docker for reproducible deployment.
- GPU training: Faster LSTM/Transformer development.
- Cloud scaling: AWS/GCP/Azure for compute elasticity.

XI. Appendices

The following appendices provide supporting used throughout the volatility forecasting engine. They serve as reference material and ensure full transparency for users and reviewers.

Feature Dictionary

This appendix describes the engineered features used by the forecasting stack.

XGBoost Feature Set

Includes:

- Price-based RV estimators
- Realised skew/kurtosis
- Compression metrics (BB, ATR%)
- VIX/VXN sensitivity
- Regime similarity features
- Macro-event proximity
- Drift features
- Relative volatility ratios (short/medium/long horizon)

LSTM Feature Set

Sequential features incorporating:

- Daily log-return windows
- Rolling realised volatility sequences
- Rolling RV ratios
- Lagged compression indicators
- Vol-momentum features

Ensemble Features

- LSTM p50 / p10 / p90
- XGB p50 / p10 / p90
- Model disagreement ($|LSTM - XGB|$)
- Ensemble spread
- Reliability Score inputs

Regime Scoring Features

- Mahalanobis drift
 - Regime similarity
 - Rolling cross-ticker volatility structure
 - Macro/vol-event distance metrics
-

Event Classification Glossary

Definitions of the event categories used by the event engine:

Macro Events

- CPI / PPI - inflation releases with high volatility impact
- NFP - Non-Farm Payrolls
- FOMC - rate decision, statement, press conference

Volatility Structure Events

- OPEX - options expiry
- VIX Expiry - futures/settlement-driven distortions
- ETF Rebalancing - large flow events affecting vol structure

Each event type includes an associated pressure score and blackout logic used by the signal engine.